Multimodal Evidence

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach evidence roles before source fusion.

Supply complete descriptions, preserve missing information and assess the selected hidden-content task before feedback.

Keep one habit: Ask what each source actually reports, what remains unavailable and which location/time assumptions support a total.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Keep visible **evidence**, hidden contents and caption claims separate

Two 45-minute sessions. A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can combine **evidence** from modalities without treating hidden content as visible.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

- Case exposure check: this source example appears in ai-2/26, ai-3/12, ai-3/13. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

- Lesson and workbook; paper and pencil

- Calculator and optional local Python 3 for arithmetic; no live API or private data

- Prepare the fresh retry separately and withhold its solution until after collection

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Speech and Audio Systems. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

Reference example, not the only acceptable wording: Teacher explanation: A multimodal pipeline may transcribe speech, inspect an image and generate a response.

Errors at one stage can propagate even when later stages run correctly. Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

Compare **conflict**ing **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information.

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of image generation and provenance

Reference example, not the only acceptable wording: Teacher explanation: An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Speech and Audio Systems

Image Generation Models

Iterative Prompting

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A caption says four seedlings; the image shows three and an opaque box. Is a fourth seedling disproved? Write an **evidence**-aware response.

Concept to explain

Multiple modalities help only when their information is aligned and relevant.

Term	Meaning
modality	A form of information, such as text, image, audio or video.
alignment	How well a system's behaviour matches the intended goals, constraints and values in a stated setting.
fusion	Combining information from multiple sources or modalities.
conflict	A disagreement between instructions, sources, labels or requirements.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach aligned source use and visible-versus-hidden reasoning

> Do not treat a described input as an actual inspected image.

State the current evidence checks

Current focus: align IDs/time/scope, attribute each reported observation, preserve hidden counts and evaluate caption claims separately. Desk description supports at least3,not exact3or5. Opening the box supplies2more only in the changed case; with no other locations, total5is then supported. Keep the older speech/inventory case as review.

Evidence question	What to record
Identity	Do picture, caption and question concern the same object or event?
Time and scope	Do they describe the same occasion, frame and counted locations?
Direct content	What does each supplied source explicitly report?
Hidden or missing	What cannot be seen or is not supplied?
Supported claim	What follows without treating a caption as a new observation?

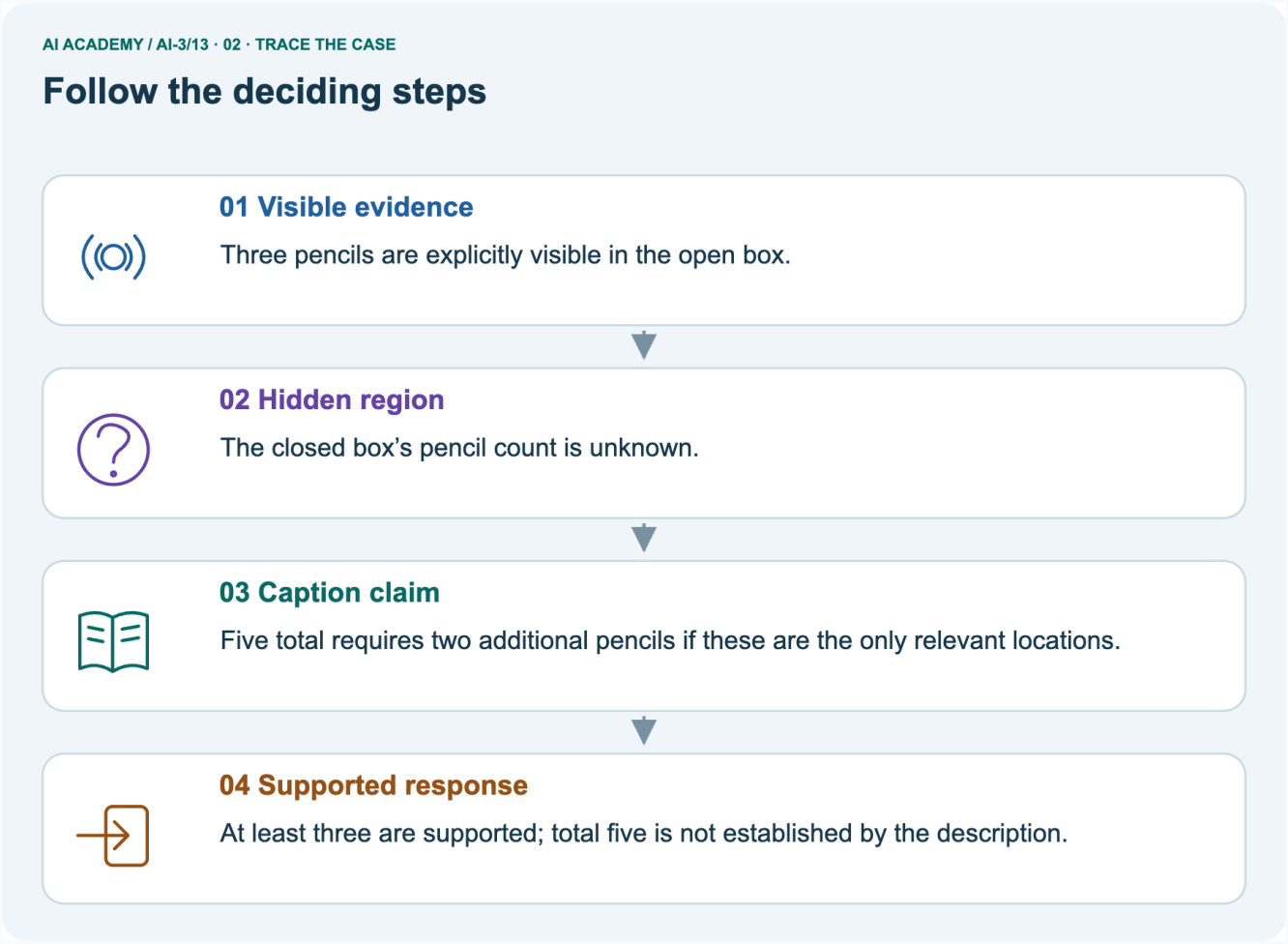
Read the complete desk case

A fictional Study Buddy audits a desk description and a separate caption.

Description: one open transparent box visibly contains three pencils; one closed opaque box hides its contents.

Caption: “There are five pencils on this desk.” No other **evidence** reveals the closed box.

Trace the lower bound and unverified caption



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Visible evidence	Three pencils are explicitly visible in the open box.
2. Hidden region	The closed box’s pencil count is unknown.
3. Caption claim	Five total requires two additional pencils if these are the only relevant locations.
4. Supported response	At least three are supported; total five is not established by the description.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A caption is another claim, not a way of seeing through an opaque object. Keep observable information separate from inferences.

The visible three give a lower bound under the stated desk scope; there is no supplied maximum for the closed box. Do not replace unknown contents with zero merely because none are visible.

Predict the new verified observation

Change: Open the closed box and a verified **observation** shows two pencils inside, with no other desk locations.

Prediction: The total is now supported as five.

Reason: 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

State description and scope limits

The conclusion depends on the supplied description's accuracy; no actual image inspection or exhaustive room inventory was performed.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Image shows leaf spots; text asks about watering history.

2. Each **modality** supplies different relevant **evidence**.

3. If photo and caption describe different plants, **fusion** is unsafe.

4. Check **alignment**, source, and uncertainty before combining.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

An image shows three plants; caption claims a fourth outside the frame.

What can the image establish?

Exact explanation to model

Three visible plants. It cannot settle an off-frame claim. Attribute claims and preserve missing **evidence**.

Explain the added worked example

A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately. Compare **conflict**ing **evidence** without assuming one **modality** always wins.

Ask for clarification when the task needs unresolved information.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Modality	A form of information such as audio, image or text
Transcription	A text representation of speech
Error propagation	An earlier error affecting later processing

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

There is an unresolved quantity **conflict** and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

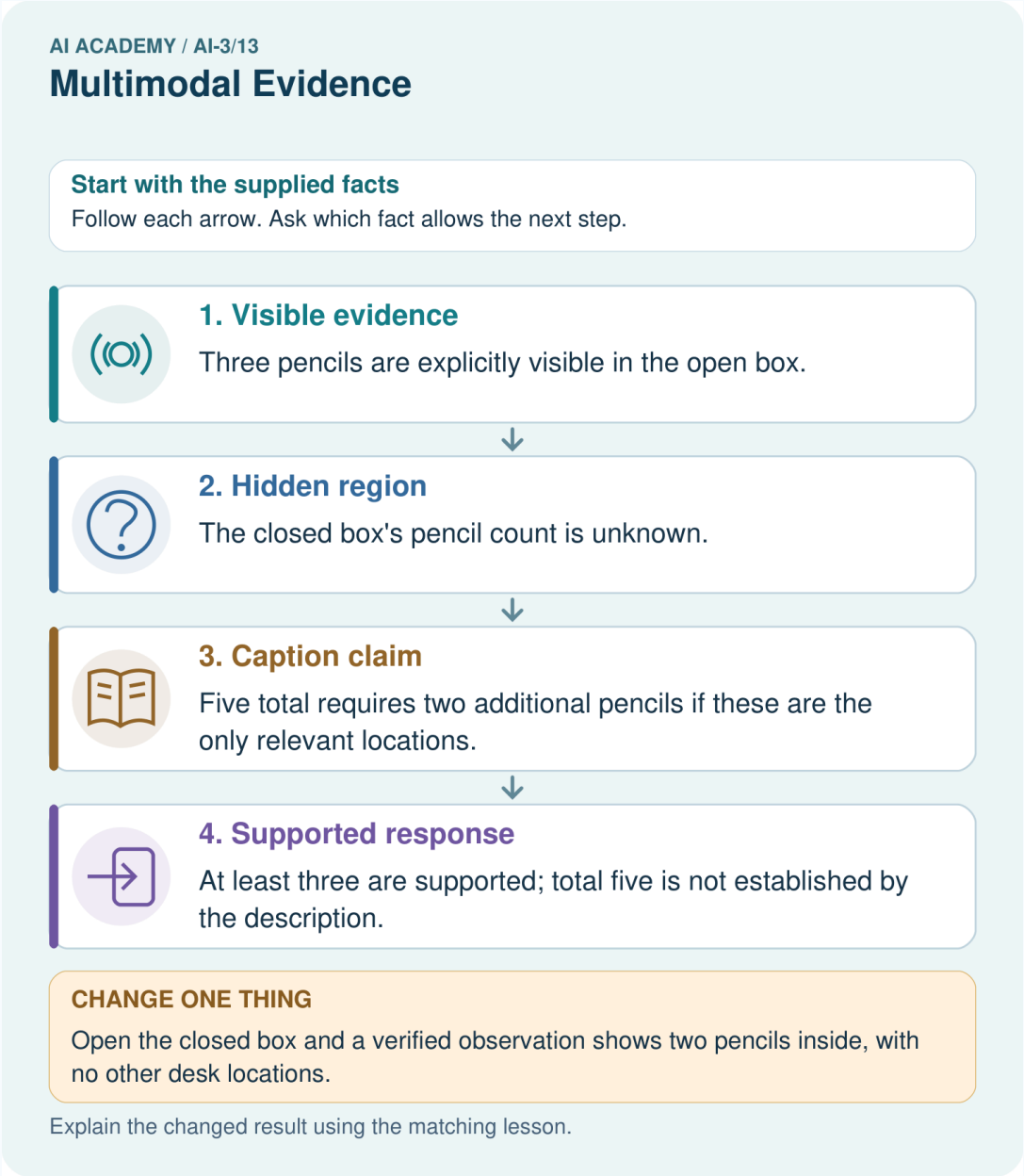
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Multimodal Evidence

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide evidence separation, then remove help

> Collect the source-role reasoning before showing feedback.

Current completion task and key

From the supplied desk description and caption, separate direct reported observations, unknown contents and unverified claims. Explain why the supported count is a lower bound rather than an exact total. Then use the supplied changed case: opening the box reveals two pencils and there are no other desk locations.

Initially the description reports three visible pencils; the hidden count is unknown and the caption’s five is unverified. At least three is supported; neither exact total is established. With the supplied verified observation of two hidden pencils and a complete-location assumption,3+2=5is supported. Do not treat the changed observation as information available before opening.

Current individual evidence check

A caption says five pencils, but the supplied description shows only three and leaves an opaque box unobserved. Does the caption establish the hidden count or does the view disprove five? Explain.

Expected correction: neither the caption nor the incomplete view establishes the hidden quantity. Preserve the visible lower bound and seek aligned additional evidence.

Pictorial changed-case reasoning

Why were both “exactly three” and “exactly five” unsupported before opening the box?

Three were visible, but hidden contents were unknown. The closed box could contain additional pencils, and none had yet been observed.

Changed-case reasoning: The total is now supported as five. 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Demonstrate and guide practice

Board sequence

Write the given facts: An image shows three plants; caption claims a fourth outside the frame.

Write the question to explain: What can the image establish?

Reveal after **prediction**s: Three visible plants. It cannot settle an off-frame claim. Attribute claims and preserve missing **evidence**.

Use the earlier changed question as optional later practice. Reserve Evaluate a new hidden-content description independently for the first independent attempt before hints or feedback.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Guided: There is an unresolved quantity **conflict** and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.
</details>

Qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion task: Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: There is an unresolved quantity **conflict** and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Speech pipelines and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied evidence-pair audit

> Keep current source-alignment reasoning separate from old speech-stage review.

Current cards and answer guidance

Desk record initially supports at least three pencils, not an exact total. The caption does not reveal the opaque box and the view does not disprove five. With two newly verified pencils and complete-location scope, total five is supported. Pair B misaligns plant IDs; C preserves a12/21report conflict; D lacks requested history. These are written descriptions, not actual image tests.

Independent plant task supports two visible plants and an unknown total; do not infer one plant per pot or a hidden zero.

Predict, attribute and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the four complete written image/text **evidence**-pair cards. Check identity,time,scope,visible content and missing history; preserve **conflict**s or ask for the relevant source. These are described fictional inputs, not actual image inspection, model results or plant diagnosis.
4. What did the hardest test teach you that the normal test did not?

Keep evidence and limits

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Current misconception support

Current misconceptions: hidden means empty—unknown is not zero; a caption makes a hidden fact visible—attribute it as a claim; matching topic implies same entity—check IDs/time/scope; a plant photo supplies past history—ask for the relevant record; equal source counts settle all conflicts—inspect meanings and authority for the actual task.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Evaluate a new hidden-content description independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected visible-plants description

> Reveal this key after the first attempt; older speech/authority questions are review.

Selected independent question and checked key

Selected case: A description reports two visible plants and a third opaque pot whose contents cannot be seen. A caption says “three plants”. What is supported?

Two visible plants are supported; total plant count is unknown. The opaque pot does not establish either presence or absence of another plant. Do not infer a total or an upper bound without more **evidence**.

Judge reasoning and scope

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Current workbook guidance

Current W1: the description reports three visible pencils, the closed box is unobserved, and the caption claims five; align object, time and scope. W2: the initial lower bound is three; two newly verified pencils plus complete-location scope support five only later. W3: the caption does not reveal hidden contents, and the incomplete view does not refute five.

W5: require a new aligned observation and explicit scope, with a predicted interpretation rather than an invented outcome. Keep the selected visible-plants case independent.

▼ Review · Earlier speech questions and keys

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately. Compare **conflict**ing **evidence** without assuming one **modality** always wins.

Ask for clarification when the task needs unresolved information.

W2 Trace the worked case

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: There is an unresolved quantity **conflict** and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

W3 Audit the limit

A report omits this qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

Teacher explanation: Report F from the authoritative approval, noting the transcript **conflict** if relevant. This resolves the task authority, not what the original speaker actually said.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/**conflict**ing/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Three visible plants. It cannot settle an off-frame claim. Attribute claims and preserve missing **evidence**.

Additional transfer practice question

A chart image shows 12 plants; its caption says 21. What should a multimodal assistant report before drawing a conclusion?

Reasoned practice answer

Report the disagreement and check the underlying source or a clearer image. Do not silently choose one channel or invent agreement.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Agreement can be assumed whenever image and text appear together.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

State that three seedlings are visible and the box contents cannot be determined. The image neither confirms nor disproves a hidden fourth seedling.

Attribute the four-seedling claim to the caption and request another view if the count matters.

Explanations for the additional workbook examples

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Accept diagrams plus questions only when the source and topic align with the learner's task.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Label Study Buddy claims by their visual, audio or text **evidence** and retain uncertainty when modalities omit information.

Support while preserving the reasoning goal

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or **conflict**ing source and validate claim by claim with a fallback.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Selected independent task: two visible plants are supported; total unknown. An opaque pot establishes neither another plant nor an empty pot, and no upper bound is supplied. Fresh retry: four visible counters beside a sealed bag;caption says six. Require at least four,unknown bag contents,and no verified total until aligned observation or complete source evidence is supplied. Record support before feedback. The earlier roomE/F authority exercise remains additional practice.

Plan the delayed check

Suggested return points in this level: ai-3/14, ai-3/16, ai-3/19. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 14: Hallucination Failure Types. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week14: Hallucination Failure Types. Carry forward source attribution, aligned **evidence** and the difference between an unsupported claim and a contradicted claim.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Verified Study Buddy

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Practice and assessment reminders

Before class, solve the supported practice: Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: There is an unresolved quantity **conflict** and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Check every learner before discussion: Use the worked case for Speech pipelines and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Open the closed box and a verified **observation** shows two pencils inside, with no other desk locations. Require a **prediction** and a reason before comparing. Expected result: The total is now supported as five.

Reason: 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

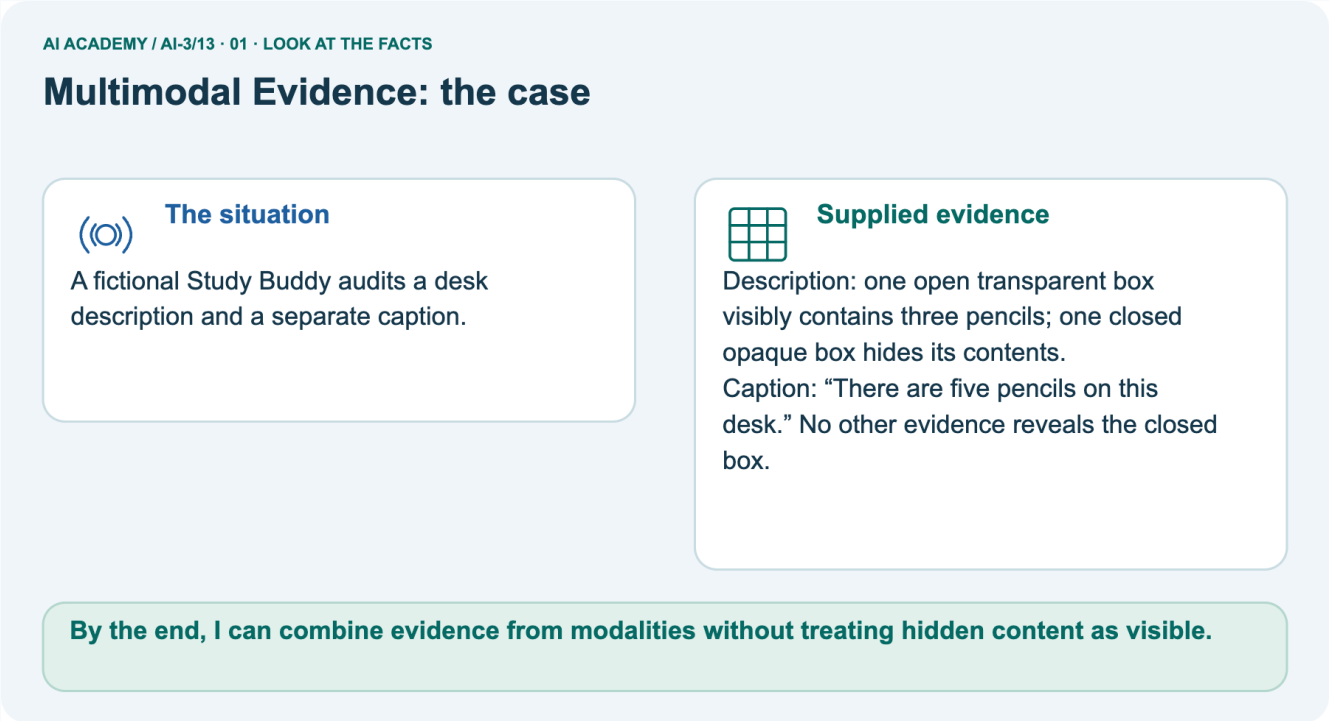
▼ Reference: teach with the six pictures

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Multimodal Evidence: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy audits a desk description and a separate caption. Supplied facts: Description: one open transparent box visibly contains three pencils; one closed opaque box hides its contents.

Caption: “There are five pencils on this desk.” No other **evidence** reveals the closed box.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



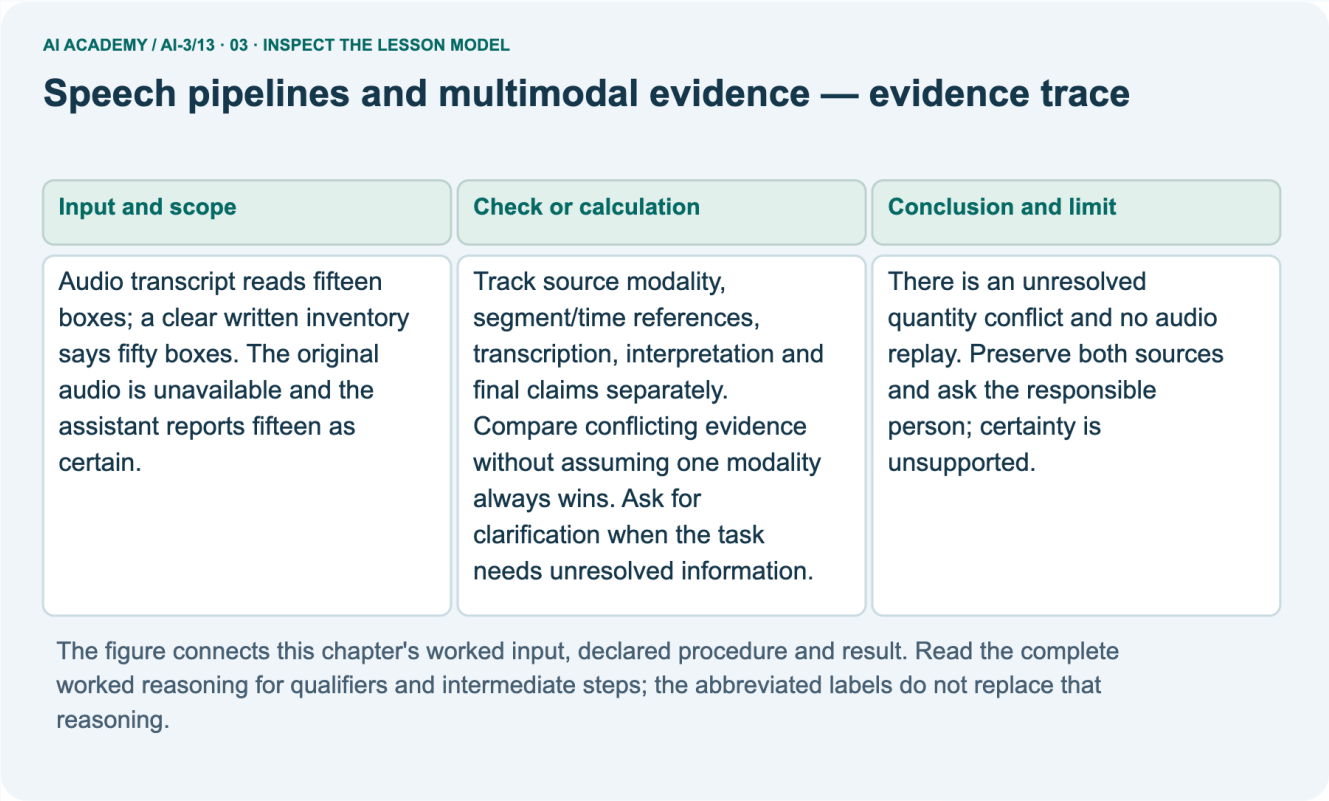
Follow the labelled picture with the matching explanation. Open the vector diagram

Visible **evidence**: Three pencils are explicitly visible in the open box. → Hidden region: The closed box's pencil count is unknown. → Caption claim: Five total requires two additional pencils if these are the only relevant locations. → Supported response: At least three are supported; total five is not established by the description.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech pipelines and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

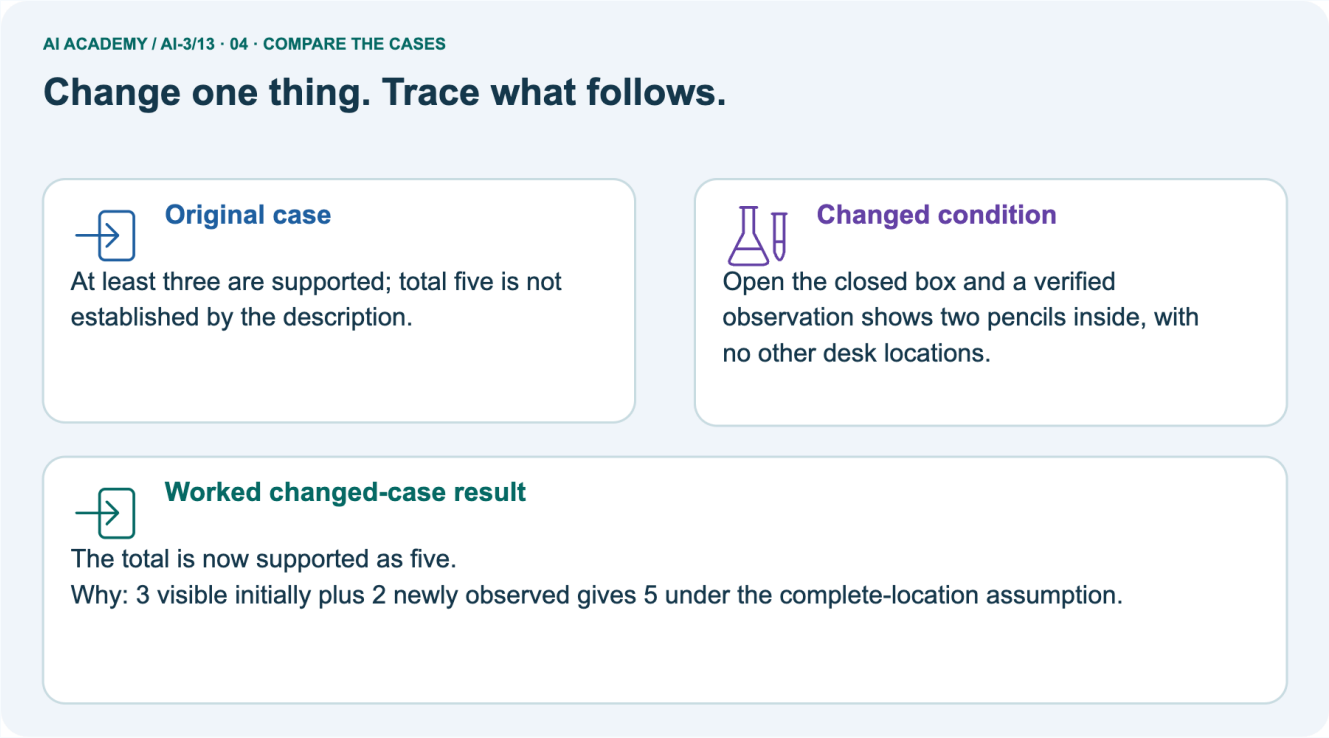
Compare **conflict**ing **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity **conflict** and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At least three are supported; total five is not established by the description. Changed condition: Open the closed box and a verified **observation** shows two pencils inside, with no other desk locations.

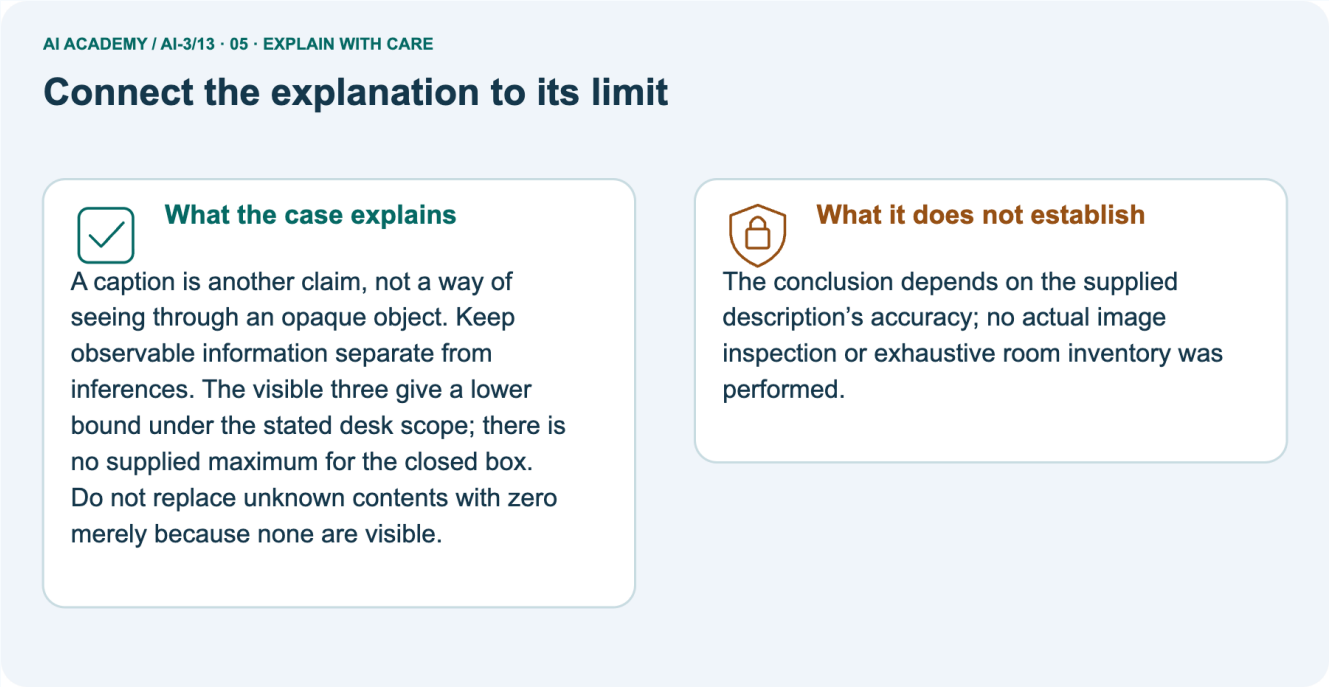
Expected result: The total is now supported as five. Reason: 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A caption is another claim, not a way of seeing through an opaque object. Keep observable information separate from inferences.

The visible three give a lower bound under the stated desk scope; there is no supplied maximum for the closed box. Do not replace unknown contents with zero merely because none are visible.

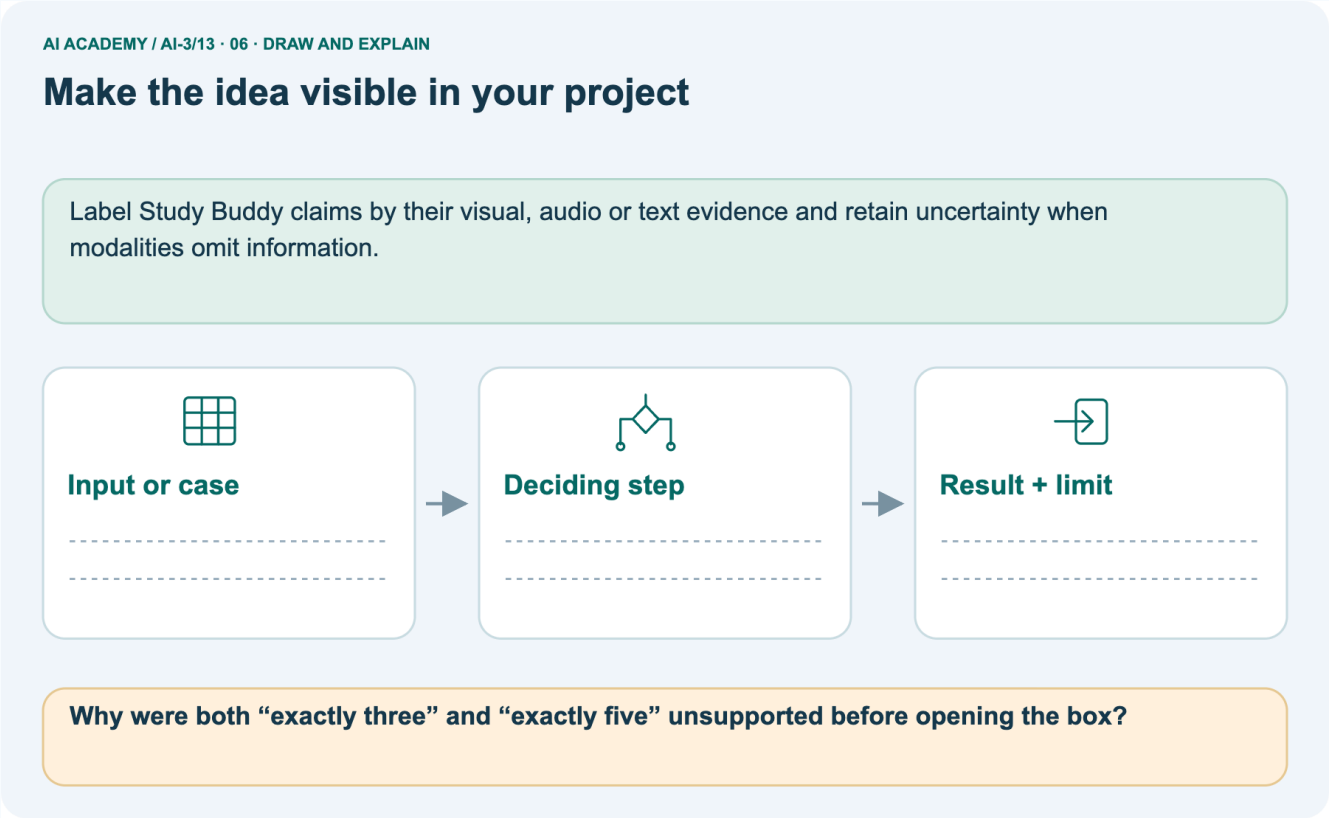
Limit: The conclusion depends on the supplied description's accuracy; no actual image inspection or exhaustive room inventory was performed.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: The conclusion depends on the supplied description's accuracy; no actual image inspection or exhaustive room inventory was performed.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Label Study Buddy claims by their visual, audio or text **evidence** and retain uncertainty when modalities omit information. Draw your **input**, deciding step, result and limit.

Supported question: Why were both “exactly three” and “exactly five” unsupported before opening the box?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Three were visible, but hidden contents were unknown. The closed box could contain additional pencils, and none had yet been observed.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Multimodal Evidence: the case

02. Follow the deciding steps

03. Speech pipelines and multimodal evidence — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.